

Part 1: Fast Forward Merge

```
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git init
```

Initialized empty Git repository in /home/monishkanna/Assignment3/.git/

```
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ touch sample.txt
```

```
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git add .
```

```
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git status
```

On branch master

No commits yet

Changes to be committed:

(use "git rm --cached <file>..." to unstage)

new file: sample.txt

```
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git commit -m "for fast forward merge in main"
```

[master (root-commit) 60294a3] for fast forward merge in main

1 file changed, 0 insertions(+), 0 deletions(-)

create mode 100644 sample.txt

```
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git branch ffm
```

```
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git switch ffm
```

Switched to branch 'ffm'

```
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ echo "Hello Guys...!!!" >> sample.txt
```

```
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git commit -a -m "for fast forward merge in ffm"
```

[ffm c8de150] for fast forward merge in ffm

1 file changed, 1 insertion(+)

```
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git branch
```

* ffm

master

```
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git switch master
```

Switched to branch 'master'

```
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git merge ffm
Updating 60294a3..c8de150
Fast-forward
 sample.txt | 1 +
 1 file changed, 1 insertion(+)
```

Part 2: Merge Commit (No Conflicts)

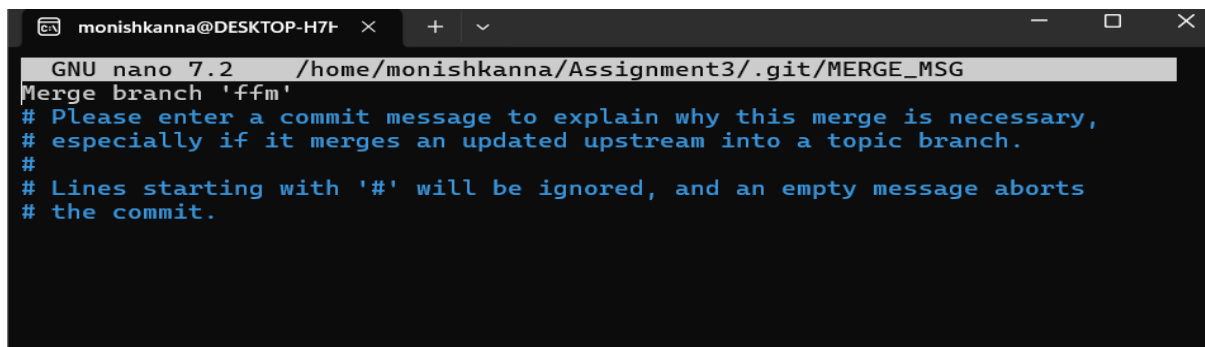
```
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git switch ffm
```

Switched to branch 'ffm'

```

monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ echo "Good Evening!" >> sample.txt
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git commit -a -m "for merge commit in ffm"
[ffm 883147e] for merge commit in ffm
1 file changed, 1 insertion(+)
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git switch master
Switched to branch 'master'
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git branch
  ffm
* master
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ touch sample2.txt
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ echo "Welcome!" > sample2.txt
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git add .
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git commit -m "for merge commit in main"
[master b88189a] for merge commit in main
1 file changed, 1 insertion(+)
create mode 100644 sample2.txt
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git merge ffm
Merge made by the 'ort' strategy.
sample.txt | 1 +
1 file changed, 1 insertion(+)

```



```

monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ nano .git/MERGE_MSG
GNU nano 7.2 /home/monishkanna/Assignment3/.git/MERGE_MSG
Merge branch 'ffm'
# Please enter a commit message to explain why this merge is necessary,
# especially if it merges an updated upstream into a topic branch.
#
# Lines starting with '#' will be ignored, and an empty message aborts
# the commit.

```

Part 3: Merge Conflict

```

monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git switch ffm
Switched to branch 'ffm'
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git switch master
Switched to branch 'master'
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git switch ffm
Switched: command not found
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ echo "change by ffm branch" >> sample.txt
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ git commit -a -m "for merge conflict in ffm branch"

```

[ffm d101f80] for merge conflict in ffm branch

1 file changed, 1 insertion(+)

monishkanna@DESKTOP-H7HK4MK:~/Assignment3\$ git switch master

Switched to branch 'master'

monishkanna@DESKTOP-H7HK4MK:~/Assignment3\$ echo "change by main branch" >> sample.txt

monishkanna@DESKTOP-H7HK4MK:~/Assignment3\$ git add .

monishkanna@DESKTOP-H7HK4MK:~/Assignment3\$ git status

On branch master

Changes to be committed:

(use "git restore --staged <file>..." to unstage)

modified: sample.txt

monishkanna@DESKTOP-H7HK4MK:~/Assignment3\$ git commit -m "for merge conflict in main"

[master e762294] for merge conflict in main

1 file changed, 1 insertion(+)

monishkanna@DESKTOP-H7HK4MK:~/Assignment3\$ git merge ffm

Auto-merging sample.txt

CONFLICT (content): Merge conflict in sample.txt

Automatic merge failed; fix conflicts and then commit the result.

monishkanna@DESKTOP-H7HK4MK:~/Assignment3\$ git add sample.txt

monishkanna@DESKTOP-H7HK4MK:~/Assignment3\$ git commit -m "resolved merge conflict"

[master 4af2fe8] resolved merge conflict

monishkanna@DESKTOP-H7HK4MK:~/Assignment3\$ cat sample.txt

Hello Guys...git switch ffm!

Good Evening!

<<<<<< HEAD

change by main branch

=====

change by ffm branch

>>>>>> ffm

```
monishkanna@DESKTOP-H7HK4MK:~/Assignment3$ cat sample.txt
Hello Guys...git switch ffm!
Good Evening!
<<<<<< HEAD
change by main branch
=====
change by ffm branch
>>>>>> ffm
```